Naïve Bayes Ant Colony Optimization for Experimental Design

Matteo Borrotti and Irene Poli

Abstract. In a large number of experimental problems the high dimensionality of the search space and economical constraints can severely limit the number of experiment points that can be tested. Under this constraints, optimization techniques perform poorly in particular when little a priori knowledge is available. In this work we investigate the possibility of combining approaches from advanced statistics and optimization algorithms to effectively explore a combinatorial search space sampling a limited number of experimental points. To this purpose we propose the Naïve Bayes Ant Colony Optimization (NACO) procedure. We tested its performance in a simulation study.

Keywords: Ant colony algorithm, combinatorial cptimization, naïve Bayes classifier.

1 Introduction

In this work we address the problem of developing a novel approach for combinatorial optimization in the context of experimental design. This approach should also improve the exploration of high dimensional spaces characterized by many interactions between variables.

More precisely we are interested in development of a new approaches for effectively exploring enzyme sequence space to improve or redesign enzyme

Matteo Borrotti · Irene Poli
European Centre for Living Technology (ECLT), San Marco 2940, Venice

Matteo Borrotti · Irene Poli
Department of Enviromental Science, Informatics and Statistics,
University Cá Foscari of Venice, Dorsoduro 2137, Venice
e-mail: {matteo.borrotti,irenpoli}@unive.it

R. Kruse et al. (Eds.): Synergies of Soft Computing and Statistics, AISC 190, pp. 489–497.
springerlink.com © Springer-Verlag Berlin Heidelberg 2013

functionality. Similar problems are presented in [1, 2]. In these works we designed a library of 95 different amino-acid sequences (*i.e.* words), that are subsequently assembled to yield a full-length enzyme (*i.e.* string) of length 4. To this regard, we can consider an enzyme as a string composed by 4 words. For each position in the string we can select an element from the set of 95 words, with repetition. The words are non-ordered discrete elements. The ultimate aim is to find the "most informative" string in according with a specific function.

Starting from this application, this work endeavours to define a new approach within statistical Design of Experiments for optimization based on Evolutionary Model Based Experimental Design [4]. In our case we have developed the Naïve Bayes Ant Colony Optimization (NACO) approach. NACO is a mixed approach that combines different methods from statistics and computer science. More precisely: (i) Naïve Bayes Classifier [6]; and (ii) Ant Colony Optimization [3], specifically $\mathcal{MAX} - \mathcal{MIN}$ Ant System [7]. In the next sections a brief description of the two approaches is given.

Our strategy identifies which elements mostly affect the response of the systems and then it adopts this information to help the metaheuristc algorithm in choosing the next set of candidate solutions.

2 Naïve Bayes Classifier

The Naïve Bayes Classifier [6] is a classification procedure based on the Bayes' rule. It assumes that the set of variables $X_1, ..., X_n$, we consider for classification, are all conditionally independent of one another, given the response Y.

We describe the conditional probability of $X = \langle X_1, ..., X_n \rangle$ on Y as:

$$P(X_1, X_2, \ldots, X_n \mid Y) = \prod_{i=1}^{n} P(X_i \mid Y)$$

This equation follows directly from the definition of conditional independence. Starting from this point, it is possible to understand how the Naïve Bayes Approach works. Assuming in fact that Y is any discrete valued variable, and the attributes $X_1, \ldots, X_n$ are any discrete or real valued variables, the goal of Naïve Bayes method is to train a classifier that will output the probability distribution over possible values of Y, for each new instance X that we want to classify.

The probability that Y will take on its k-th possible value, according to the Bayes' Rule, is

$$P(Y = y_k \mid X_1, \ldots X_n) = \frac{P(Y = y_k)P(X_1, \ldots, X_n \mid Y = y_k)}{\sum_j P(Y = y_j)P(X_1, \ldots, X_n \mid Y = y_j)}$$

where the sum is taken over all possible value y_j of Y. Assuming that X_i are conditionally independent given Y, we can write

$$P(Y = y_k \mid X_1, \ldots X_n) = \frac{P(Y = y_k) \prod_i P(X_1, \ldots, X_n \mid Y = y_k)}{\sum_j P(Y = y_j) \prod_i P(X_i \mid Y = y_j)} \quad (1)$$

Equation (1) is the base for the Naïve Bayes Classifier. Given a new instance $X^{new} = \langle X_1, ..., X_n \rangle$, it is possible to calculate the probability that Y will take on any given value. If we want to know the most probable value of Y, we obtain the Naïve Bayes' Rule:

$$Y \leftarrow arg \max_{y_k} \frac{P(Y = y_k) \prod_i P(X_1, \ldots, X_n \mid Y = y_k)}{\sum_j P(Y = y_j) \prod_i P(X_i \mid Y = y_j)}$$

This approach reduces the complexity for learning Bayesian classifier by making a conditional independence assumption that dramatically reduces the number of parameters to be estimated when modeling $P(X \mid Y)$.

3 Ant Colony Optimization (ACO)

Ant Colony Optimization (ACO) [3] is a population-based, general-purpose stochastic search technique for the solution of difficult combinatorial problems, which is inspired by the pheromone trail laying and following foraging behaviour of some real ant species.

In ACO, each "ant" builds a solution starting from an initial state selected according to some problem dependent criteria. A solution is expressed as minimum cost (shortest) path through the states of the problem in accordance with the problem's constraints. A single ant is able to build a solution but only the cooperation among all the agents of the colony, concurrently building different solutions, is able to find high quality solutions.

In the algorithm an environment is simulated by a graph composed of a set N of states, representing nodes, and a set E of arcs fully connecting the nodes N. Let d_{ij} be the length of the arc $(i, j) \in E$, that is the distance between nodes i and j, with $i, j \in N$. The aim of the optimization in ACO is to find on the graph $G = (N, E)$ the minimal length path connecting nest to the source.

ACO uses two different types of information: pheromone and *a priori* problem-specific information (heuristic values). The combination of available pheromone and heuristic values defines *ant-decision tables*, that are, probabilistic tables used by the ants' decision policy to direct their search towards the most interesting regions of the search space.

The ant-decision table $A_i = [a_j^i(t)]_{|N_i|}$ of node i is obtained by the composition of the pheromone trail values with heuristic values as follow:

$$a_{ij} = \frac{[\tau_{ij}(t)]^\alpha [\eta_{ij}]^\beta}{\sum_{l \in N_i} [\tau_{ij}(t)]^\alpha [\eta_{ij}]^\beta} \qquad \forall j \in N_i \tag{2}$$

where $\tau_{ij}(t)$ is the amount of pheromone trail on arc (i, j) at time t; $\eta_{i,j} = 1/d_{ij}$ is the heuristic value to move from node i to node j; N_i is the set of neighbours of node i; and α and β are two parameters that control the relative weight of pheromone trail and heuristic information.

The probability with which an ant k chooses to go from node i to node $j \in N_i^k$ while building its tour at the t-th algorithm iteration, is:

$$p_{ij}^k(t) = \frac{a_{ij}(t)}{\sum_{i \in N_i^k} a_{ij}(t)} \tag{3}$$

where $N_i^k \subseteq N_i$ is the set of nodes in the neighbourhood of node i that ant k has not visited yet.

After all ants have completed their tour, pheromone evaporation on all arcs is applied in according with the pheromone trail decay coefficient (or evaporation factor) ρ, $\rho \in (0, 1]$.

In our case, we apply the $\mathcal{MAX} - \mathcal{MIN}$ Ant System ($\mathcal{MM}$AS), proposed by Stützle and Hoos in [7] because it is demonstrated that $\mathcal{MM}$AS is able to reach a strong exploitation of the search space by adding pheromone only to the best solution during the pheromone trail update. Moreover they applied a simple method for limiting the strength of the pheromone trails that effectively avoids premature convergence of the search.

4 The NACO Approach

We introduce a novel approach for optimization called Naïve Bayes Ant Colony Optimization (NACO) by combining Ant Colony Optimization technique and Naïve Bayes Classifier. NACO extracts the information from the data using the Naïve Bayes Approach and explores the search space by the ACO algorithm. At the same time, the most informative variables and interactions are identified.

In our problem we are considering a specific problem caracterized by a discrete search space where a solution is a string composed by 4 words. The words are selected from a set of 95 discrete elements with repetition. We need to reformulate the problem as a path search problem to apply ACO algorithm. For this purpose, we create a graph where each node represents a specific word. A solution is a path with length 4 composed of 4 nodes connected by 3 arcs, as shown in Fig. 1. In the biological application [1], each node corresponds to an amino-acid sequence from the initial dataset

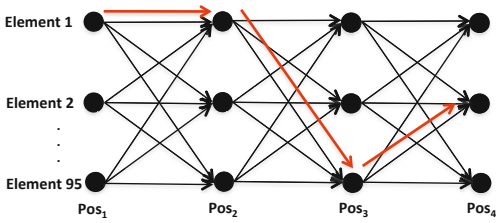

Fig. 1 A new representation of the graph where ants move. A solution is a path composed by 4 nodes and 3 arcs.

and an arc to the connection between amino-acid sequence i in position k and amino-acid sequence j in position $k+1$. A candidate solution is a path composed by 4 amino-acid sequences which represents a full-length enzyme. The total number of nodes is $95 \times 4 = 380$ with equal size intervals.

Candidate solutions' response is calculated and subsequently discretized in according with a certain constant threshold (γ, with $\gamma \in \mathbb{R}$) fixed by the experimenter. The candidate solutions (strings) are assigned to class 1 if the responses exceed the threshold otherwise to class 0. The Naïve Bayes Classifier is applied on each position of the sequence or string with class equal to 1. It calculates the maximum likelihood estimate for each word in each position given the training sequences or strings. We obtain a set of probability distributions, one for each position. These probability distributions are used to weigh the arcs on the ACO algorithm.

The following steps summarize the NACO approach:

1. Random generation and evaluation of an initial population (set of candidate points);
2. Identification of the *Iteration Best Solution* (best solution in the current iteration);
3. Calculation of the Naïve Bayes Classifier on the available evaluated solutions (N). At each iteration, it focuses on values of the response greater than a problem-specific threshold γ, with $\gamma \in \mathbb{R}$;
4. Updating the probabilities with which an ant k chooses to go from element i to element j using the information extracted in points 2 and 3;
5. Selection of the next population of candidate solutions using the principle of $\mathcal{MAX} - \mathcal{MIN}$ Ant System;
6. Evaluation of the new set of candidate solutions and inclusion of the new set in the set of solutions that has already been evaluated;
7. If stop criterion is reached, then stop. Otherwise repeat points from 2 to 6;

Fig. 2 describes point number 4. At iteration t, agents move over the graph according to the best paths identified in the previous steps (Fig. 2 (a)).

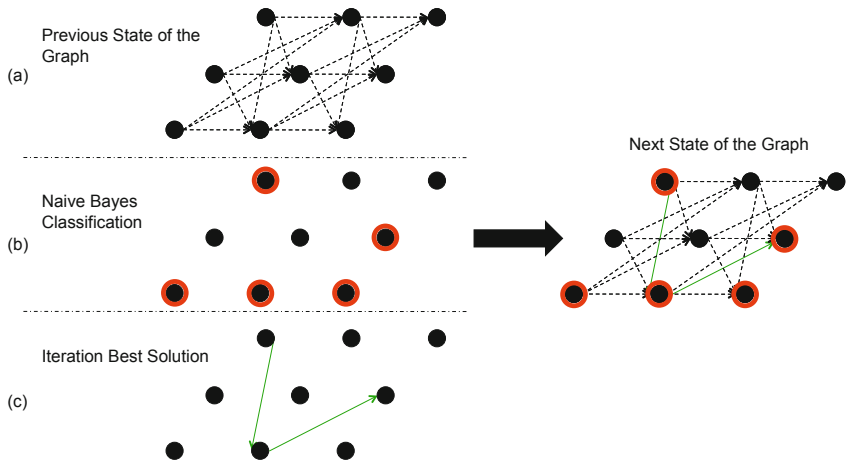

Fig. 2 Updating Phase of the Naïve Bayes Ant Colony Optimization

Following candidate solution evaluation, the *Iteration Best Solution* is identified and the corresponding pheromone path is updated (Fig. 2 (c)). At this point, using the Naïve Bayes Classifier the best variables are identified, namely those that are anticipated to yield a fitness value higher than the chosen fitness treshold γ (Fig. 2 (b)). For any given arc connecting variable (node) i with variable j, the weight λ_{ij} is changed according to the Naïve Bayes Classifier. The set of $\{\lambda_{ij}\}$ is called *Naïve Information*. Now, the ant-decision table $A_i = [a^i_j(t)]_{|N_i|}$ of node i will be obtained by the composition of the pheromone trail values with heuristic values and with *Naïve Information* as follows:

$$a_{ij} = \frac{[\tau_{ij}(t)]^\alpha [\eta_{ij}]^\beta [\lambda_{ij}]^\delta}{\sum_{l \in N_i} [\tau_{ij}(t)]^\alpha [\eta_{ij}]^\beta [\lambda_{ij}]^\delta]} \qquad \forall j \in N_i$$

where $\tau_{ij}(t)$ is the amount of pheromone trail on arc (i, j) at time t; $\eta_{i,j} = 1/d_{ij}$ is the heuristic value of moving from node i to node j; λ_{ij} is the *Naïve Information* on arc (i, j) at time t. N_i is the set of neighbors of node i; and α, β and δ are parameters that control the relative weight of pheromone trail, heuristic information and *Naïve Information*. The probability, which an ant k chooses to go from element i to element j, is then calculated according to Equation (3).

Generally, NACO will extract information from few data and it will individuate the best connection between variables.

5 Monte Carlo Simulations

In this work we develop simulative studies with the aim of testing the performance of the NACO approach. In these simulations the experiments are generated by the model $y = \varphi_h(\mathbf{x}) + \epsilon$, $\epsilon \sim N(0, \sigma_h^2)$, where $h = 1, 2$ denotes two different response surfaces, $\varphi_1(\mathbf{x})$ and $\varphi_2(\mathbf{x})$, taken from [2]. $\varphi_1(\mathbf{x})$ is called Polynomial Regression Model (PRM) and it represents a fitness landscape dominated by strong interactions, which occurs when the effect of one variables depends on the presence of another. The second formal structure, $\varphi_2(\mathbf{x})$, is called Polynomial Sparse Regression Model (PSRM) and it represents the situation where only few variables highly influence the response of the system and the others are close to 0. This kind of fitness landscapes is characterized by ruggedness and local optima. $\varphi_1(\mathbf{x})$ and $\varphi_2(\mathbf{x})$ simulate a search space of 95^4 experimental points.

In both simulations, we computed 50 Monte Carlo runs with the aim to maximize the deterministic function φ_h.

In Fig. 3, we compare the convergence of our method with the basic $\mathcal{MM}$AS. The parameter settings for $\mathcal{MM}$AS is: population size (N) 100, number of generations or experimental batches 50, evaporation factor (ρ) 0.96 and weight for the pheromone (β) 1. In this case no heuristic information is used. NACO includes two more parameters: δ that controls the weight of *Naïve Information* and γ that is the threshold considered by the Naïve Bayes Classifier. In our case δ and γ are equal to 2 and 80%, respectively. The parameter setting is chosen in accordance with preliminary studies [1].

Our empirical results (Fig. 3) show remarkable efficiency of the proposed method. The main difference between NACO and $\mathcal{MM}$AS is that the last

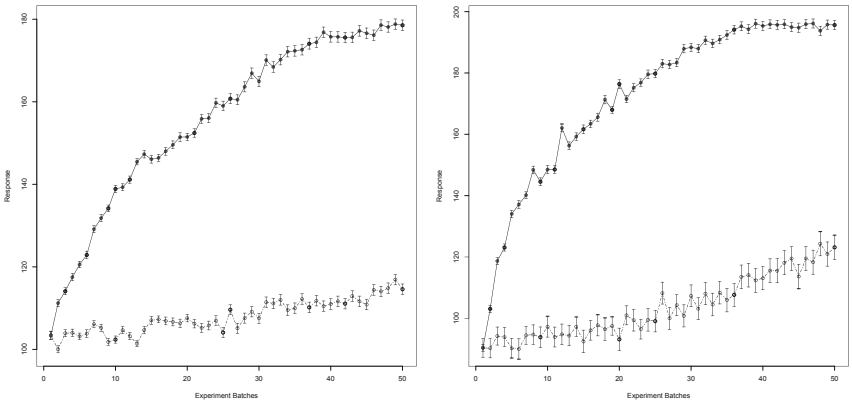

Fig. 3 Convergence behavior for Experiment 1 ($\varphi_1(\mathbf{x})$, left side) and Experiment 2 ($\varphi_2(\mathbf{x})$, right side) of the NACO (solid lines) and $\mathcal{MM}$AS (dashed lines) methods. The vertical bars are 95% Monte Carlo confidence intervals.

one does not benefit from the information obtained using the Naïve Bayes Classifier. Our results show that the inclusion of the Naïve Bayes Classifier boosts the convergence rate. This is visible after few iterations of the NACO algorithm and is due to the fact that the Naïve Bayes Classifier extracts useful information on the main variables that influence the response. The information extrapoleted from the Naïve Bayes Classifier compensates for the absence of heuristic information.

In term of computational time the two approaches do not show significante difference.

6 Conclusions

In this work, we have explored the possibility of tackling problems characterized by a very large experimental space combining bio-inspired algorithms with advanced statistical techniques. To achieve this aim, we have developed an algorithmic approach which combines some powerful features of known approaches: the Naïve Bayes Ant Colony Optimization (NACO).

We have shown that the Naïve Bayes Ant Colony Optimization (NACO) approach improves upon the limits of the individual techniques, enabling us to deal with very large experimental spaces. In fact, the Naïve Bayes Approach has a strong assumption and it assumes that the attributes $X_1, \ldots, X_n$ are all conditionally independent of one another, given the response Y. It has the advantage of simplifying the representation of the probability of X given Y but with the Naïve Bayes Classificator it is not possible to understand the relations between the attributes. This aspect can be extremely important in some experimental problems.

The combination of ACO and Naïve Bayes Classifier can describe the relative network between variables. ACO, in fact, is based on probabilistic matrices where the best path has a higher probability of being chosen. A path is composed of nodes and arcs. In our problem nodes can be seen as variables and an arc, connecting a variable to the next one, can be seen as the relation that exists between the two variables. Then, ACO implies the sequential relationship between variables.

At last, NACO can be used in the context of combinatorial optimization in absence of heuristc information since Naïve Bayes Classifier is used to extract information from the available data.

References

1. Borrotti, M.: An evolutionary approach to the design of experiments for combinatorial optimization with an application to enzyme engineering. PhD thesis, Department of Statistical Science, University of Bologna, Italy (2011)

2. Borrotti, M., De Lucrezia, D., Minervini, G., Poli, I.: A Model Based Ant Colony Design for the Protein Engineering Problem. In: Dorigo, M., Birattari, M., Di Caro, G.A., Doursat, R., Engelbrecht, A.P., Floreano, D., Gambardella, L.M., Groß, R., Şahin, E., Sayama, H., Stützle, T. (eds.) ANTS 2010. LNCS, vol. 6234, pp. 352–359. Springer, Heidelberg (2010)
3. Dorigo, M., Di Caro, G.: The ant colony optimization meta-heuristic. In: Corne, D., et al. (eds.) New Ideas Optim., pp. 11–32. McGraw-Hill, New York (1999)
4. Forlin, M., Slanzi, D., Poli, I.: Combining Probabilistic Dependency Models and Particle Swarm Optimization for Parameter Inference in Stochastic Biological Systems. In: Gaol, F.L., Nguyen, Q.V. (eds.) Proc. of the 2011 2nd International Congress CACS. AISC, vol. 145, pp. 437–444. Springer, Heidelberg (2012)
5. Kohonen, J., Talikota, S., Corander, J., Auvinen, P., Arjas, E.: A naïve bayes classifier for protein function prediction. Silico Biol. 9, 23–34 (2009)
6. Mitchell, T.M.: Machine Learning. McGraw-Hill, New York (1997)
7. Stützle, T., Hoos, H.: Max-min ant system. Future Gener. Comp. Sy. 16, 889–914 (2000)